

The angle between a face and the base

ANSWER KEY • FOR TEACHERS



Interactive version

CHECK

Check yourself

The figure shows the same pyramid: base side 4, height $SO = 3$.

Level 2 • satisfactory

1.

Which angle is the angle between the lateral edge **SB** and the base?

☐ A $\angle SMO$
☒ B $\angle SBO$ — SO is perpendicular to the base, so OB is the projection of SB , and the angle is $\angle SBO$. Here $\tan \angle SBO = \frac{SO}{OB} = \frac{3}{2\sqrt{2}}$.

☐ C $\angle BSO$
☐ D $\angle SBC$

Level 2 • satisfactory

2.

The angle between the face **SCD** and the base is measured at one point of the edge CD . Tap that point on the figure.

Answer: It is K , the midpoint of CD : $OK \perp CD$ in the base and $SK \perp CD$ in the face, so $\angle SKO$ is the angle. By symmetry it equals φ .

Level 2 • satisfactory

3.

Find the length of the apothem SM . Give the exact value or round to two decimal places.

WORKED SOLUTION

In triangle SOM : $SM^2 = SO^2 + OM^2 = 9 + 4 = 13$, so $SM = \sqrt{13} \approx 3.61$.

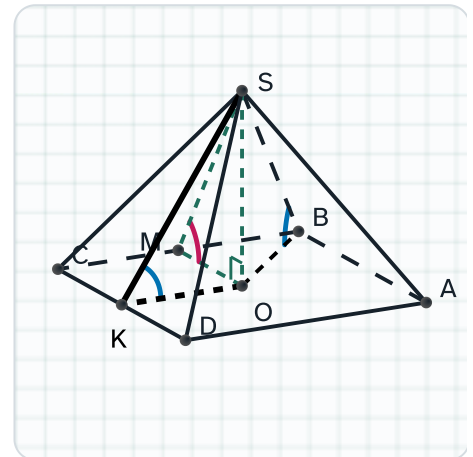
Answer: $SM = \sqrt{13} \approx 3.61$

Level 3 • basic

4.

Find the angle φ between a side face and the base, in degrees, rounded to 0.1° .

Answer: $\varphi \approx 56.3^\circ$



K is the midpoint of CD .

EXAM TASK

A regular pyramid, VBE style

Level 4 · advanced

VBE Part II style · 6 points

The base of a regular quadrilateral pyramid is a square with side 8. The angle between a side face and the base is 60° .

1. Show that the height of the pyramid is $4\sqrt{3}$.
2. Find the lateral surface area of the pyramid.
3. Find the angle between a lateral edge and the base, to 0.1° .

MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

Identifies $\angle SMO = 60^\circ$ with $OM = 4$ in the right triangle SOM. 1 pt

Shows $SO = OM \cdot \tan 60^\circ = 4\sqrt{3}$. 1 pt

Finds the apothem $SM = \frac{OM}{\cos 60^\circ} = 8$. 1 pt

Lateral surface area $4 \cdot \frac{1}{2} \cdot 8 \cdot 8 = 128$. 1 pt

Uses $\angle SBO$ with $OB = 4\sqrt{2}$. 1 pt

$\tan \angle SBO = \frac{4\sqrt{3}}{4\sqrt{2}} = \sqrt{\frac{3}{2}}$, so the angle is $\approx 50.8^\circ$. 1 pt

WORKED SOLUTION

1. Let M be the midpoint of BC and O the centre of the base. Then $OM \perp BC$ and $SM \perp BC$, so $\angle SMO = 60^\circ$ is the angle between the face SBC and the base. In the right triangle SOM, $OM = 4$, so $SO = OM \tan 60^\circ = 4\sqrt{3}$, as required.
2. $SM = \frac{OM}{\cos 60^\circ} = 8$. Each side face is a triangle with base 8 and height 8, so the lateral area is $4 \cdot \frac{1}{2} \cdot 8 \cdot 8 = 128$.
3. SO is perpendicular to the base, so the projection of SB is OB, and the angle is $\angle SBO$. $OB = \frac{1}{2} \cdot 8\sqrt{2} = 4\sqrt{2}$, so $\tan \angle SBO = \frac{4\sqrt{3}}{4\sqrt{2}} = \sqrt{1.5}$ and $\angle SBO \approx 50.8^\circ$.

TEACHER NOTES

For the teacher

Pupils can say: "I can find the angle between a side face and the base of a regular pyramid, and I can tell it apart from the angle between a lateral edge and the base."

Common mistakes and what helps

- Choosing $\angle SBO$ (edge and base) instead of $\angle SMO$ (face and base): ask "is it a line or a plane that meets the base?"
- Taking an angle inside one face, such as $\angle SBC$.
- Using OB instead of OM in triangle SOM.
- Trusting the drawing: in the exam view the angle does not appear at its true size. "Look along BC" shows why.

Key terms (EN · LT): dihedral angle · *dvisienis kampas*; regular pyramid · *taisyklingoji piramidė*; apothem · *apotema*; side face · *šoninė siena*; lateral edge · *šoninė briauna*; base · *pagrindas*; projection · *projekcija*.

Exam relevance: a regular pyramid, cone or cylinder with angles and distances was in every VBE Part II paper from 2019 to 2026 (our analysis of the published papers). “Show that” parts have the lowest success rates in NŠA item statistics; the exam task practises one.

Programme: grade IV, “Lines, planes and angles in space”: the dihedral angle, both courses (V-594 text). Achievements A1, B1, C1.